Classification of Legal Text: Replication & Extension

Mini-Project Write-up | Replicating & extending Krithika Iyer, *"Classification of Legal Text,"* Stanford CS229 (2020)

Problem Statement

Legal opinions use long, non-standard sentence structures that break typical NLP tooling. We replicate and extend a Stanford CS229 project that classifies U.S. Supreme Court opinions into their human-curated legal subject-matter category, at two granularities: **issue_area** (15 broad categories) and **issue** (260+ fine-grained codes). Beyond replication, we test whether modern pretrained embeddings beat the paper's from-scratch Doc2Vec.

Dataset

The exact dataset the original paper used: the Supreme Court Database via the **textacy** package's bundled **SupremeCourt** dataset — 8,419 opinions (1946-2016). After dropping opinions with no issue_area label and issue codes with fewer than 10 examples, 7,971 opinions remain, split 70/15/15 into train/dev/test, stratified by issue_area. Opinions run long (median ~34,000 characters); each is truncated to its first 4,000 characters before vectorization.

Approach — Five Phases

Features: (1) LDA — gensim topic model, TF-IDF, no training beyond fitting the topic distribution. (2) Doc2Vec — a small neural net, trained from scratch on our ~5,600 training documents only. (3) SBERT (all-MiniLM-L6-v2) — a transformer pretrained on 1B+ sentence pairs; used only to encode, never trained. **Classifiers:** Logistic Regression (the paper's), plus SVM, Random Forest, and Gaussian Naive Bayes.

Phase 1 replicates the paper's Doc2Vec+LR with its default settings. **Phase 2** tunes each feature extractor's main hyperparameter (LDA num_topics, Doc2Vec vector_size) and each classifier's C, selected on dev. **Phase 3** compares SVM/Random Forest/Naive Bayes against LR on the same tuned features (the paper never does this). **Phase 4** deep-tunes just the single best model from Phase 3 (SVM + Doc2Vec + issue_area) over 40 kernel/C/gamma combinations, selecting on dev *accuracy* specifically. **Phase 5** swaps Doc2Vec for SBERT — zero training for the embedding step — on the same classifiers and tasks. All hyperparameter and classifier selection uses only the dev split; test is evaluated once per final configuration.

Implementation

`src/data/` (corpus loading, splitting, labels), `src/models/` (lda_lr, doc2vec_lr, sbert_lr, classifiers), `scripts/01_replicate.py` … `05_sbert.py` (the five phases, in order), `scripts/train_best_model.py` + `predict.py` (saves the best model, classifies new text for a live demo), `scripts/evaluate_best_model.py` (confusion matrix).

Results — Progress Across All Five Phases (issue_area)

Phase	Model	Test Acc	Test Macro-F1
1 — Replicate	Doc2Vec + LR (paper defaults)	59.1%	0.463
2 — Tune features	Doc2Vec + LR (tuned vector_size, C)	62.2%	0.477
3 — Compare classifiers	Doc2Vec + SVM (default)	68.8%	0.499
4 — Deep-tune winner	Doc2Vec + SVM (poly)	68.5%	0.547
5 — Pretrained embeddings	SBERT + SVM	70.1%	0.526

Phase 4 selected on dev *accuracy* (not macro-F1, unlike earlier phases) specifically because the deployed model's accuracy is what matters in practice; even so, test accuracy held flat (68.8%→68.5%, within noise for an ~850-example test set) while macro-F1 rose (0.499→0.547) via a kernel change (RBF→poly). Phase 5's SBERT then beat every Doc2Vec variant at every feature/task/classifier combination we tried — a modern pretrained model, with zero training, outperforms a model trained from scratch on our small corpus.

Per-Category Pattern (issue_area)

LDA's macro-F1 trails its accuracy badly, defaulting to majority classes (Criminal Procedure, Economic Activity) and collapsing to near-zero F1 on minority ones (Privacy, Attorneys). Doc2Vec and SBERT both generalize far better to minority categories, since they capture context rather than just word co-occurrence. Error analysis shows the best model reads surface vocabulary, not legal reasoning: an Attorneys fee-dispute opinion dominated by bond/surety financial language was misclassified as Federal

Taxation (97% confidence), while an opinion with explicit labor-law vocabulary (“Secretary of Labor,” “union election”) was correctly classified as Unions.

Conclusions & Limitations

Pretrained embeddings beat from-scratch ones at every comparison point; classifier choice matters almost as much as features (SVM beat every other classifier tried); tuning helped in every phase, though Phase 4 traded flat accuracy for a macro-F1 gain; the coarse 15-category task is far easier than the 260+-category one everywhere we tried (best ~48% accuracy) — a data scarcity problem, not a model problem. Overall: 59.1% → 70.1% test accuracy across five phases, without ever touching the dataset. Our replicated numbers land in the same range and order as the original paper's; we also note an internal inconsistency in the paper's own results table (its printed table and prose disagree on which label granularity scored higher) and resolve it in favor of the prose. **Limitations:** single train/dev/test split (no cross-validation, so small accuracy gaps carry real uncertainty); 4,000-character truncation discards most of each opinion; fine-grained issue classification remains weak everywhere. **Future work:** a legal-domain pretrained model (Legal-BERT) and full BERT fine-tuning (vs. SBERT's frozen embeddings) are natural next steps, as the original paper itself suggested before its own attempt ran out of memory.

Full code, results, and figures: github.com/amogh-herle/ML-MINI-PROJECT